Lecture 7: Trigonometric Substitutions

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

Sem II, 2025/2026

Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Perform the integration by using a suitable trigonometric substitutions.

Introduction

Many integrals can be evaluated by using a right substitution, however, there is no general way in finding the right one unfortunately.

Integrals involving the square roots of quadratics often yield to trigonometric substitution.

For $a > 0$,

Expression in integrand	Trigonometric substitution
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$

Trigonometric Substitutions

Substitution of $x = a\sin\theta$ in $\sqrt{a^2 - x^2}$

Substitution of $x = a\tan\theta$ in $\sqrt{a^2 + x^2}$

Substitution of $x = a\sec\theta$ in $\sqrt{x^2 - a^2}$

Examples

$$\int \frac{dx}{\sqrt{4+x^2}}$$

Examples

$$\int \frac{x^2\,dx}{\sqrt{9-x^2}}$$

Examples

$$\int \frac{dx}{\sqrt{25x^2-4}},\ x > \frac{2}{5}$$

Examples

$$\int \frac{dx}{(4+x^2)^{\frac{3}{2}}}$$

Examples

$$\int_0^3 \frac{75}{(25-x^2)^{\frac{3}{2}}}\,dx$$